EDUMATCH

Predictive Modeling and Prescriptive Analytics for Early Student Retention Intervention

An end-to-end educational performance metric with local regulatory knowledge bases using machine learning and RAG architecture.

TECH STACK: Python, Pandas, NumPy, Scikit-Learn (Random Forest, K-Means, Standard Scaler), Matplotlib, Seaborn, LLMS, Streamlit

PAUL KWAKU LARBI ANDERSON

Main Objective: Mitigating Early Student Attrition

The Core Problem

Higher education institutions face severe drop-out rates during the first two semesters of undergraduate study.

Systemic Challenge

Traditional intervention models react to student failures — registration anomalies or failing grades — rather than acting preemptively.

Action Pathway

This project constructs an end-to-end predictive pipeline to identify ‘at-risk’ students from baseline demographics, socioeconomic stressors, and initial academic trajectory.

The Database: Ingestion and Exploration

Data Alignment: Portugal Dataset to German RAG Engine

Portugal Dataset (Kaggle- CSV)

4,424-row quantitative training core tracking undergraduate journeys at the Polytechnic Institute of Portalegre.

Bologna Process

Standardized ECTS credit accumulation, equivalent winter/summer semesters, and matching grading metrics enable domain transferability.

German RAG Engine (PDF to Text)

Prescriptive layer driven by the unstructured text rules of the University of Siegen General Examination Regulations (Prüfungsordnung).

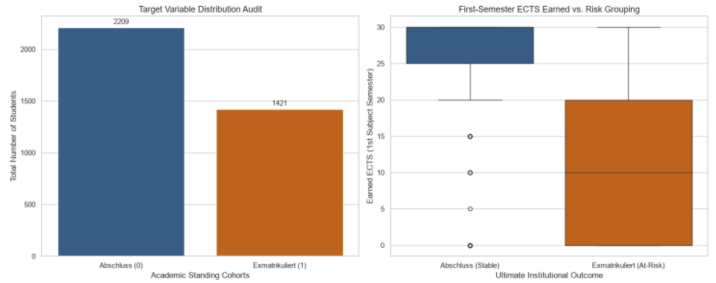
Data Harmonization Strategy

Though trained on Portuguese behavioral features and applied to German rules, both institutions operate under the European Bologna Process. ***Unified ECTS benchmarks and matching semester structures allow robust domain transferability*** between the two academic systems.

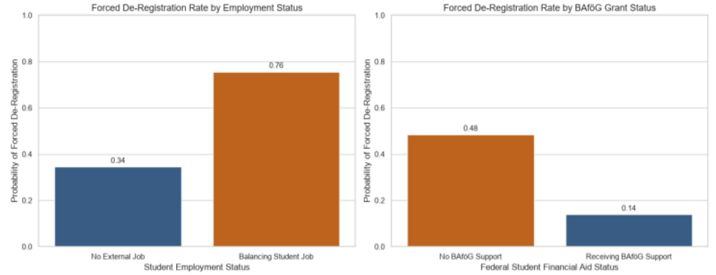
Unified Features: ECTS credit accumulation • equivalent winter/summer semester divisions • matching baseline grading metrics

Supporting Data Visualization Images

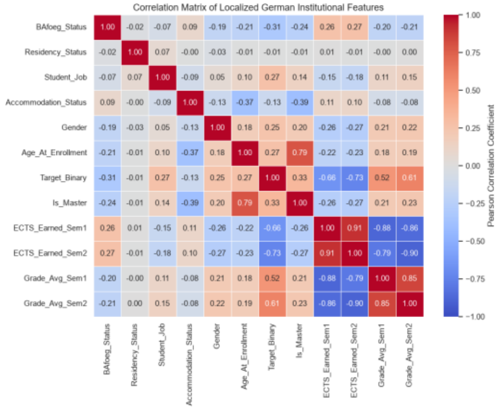
EDA, model driver, and validation charts underpinning the EduMatch pipeline



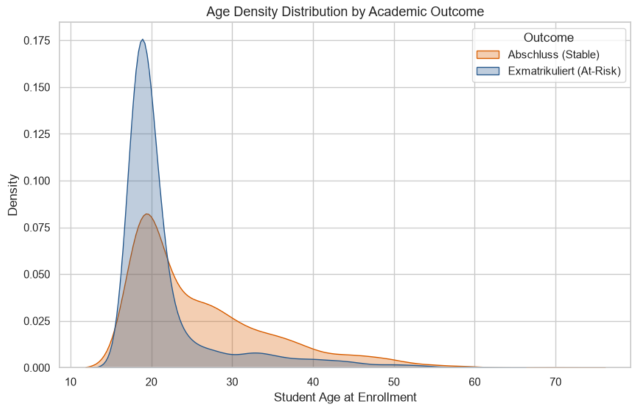
Target Variable Audit & ECTS vs. Risk Grouping



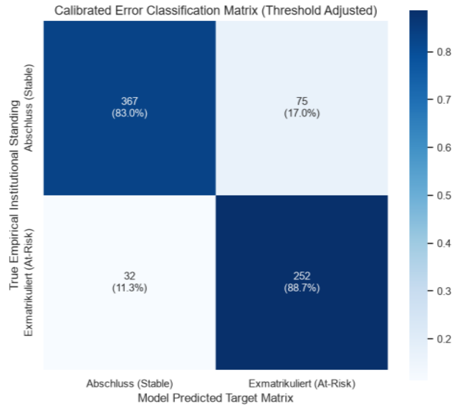
De-Registration Rate by Employment & BAföG Status



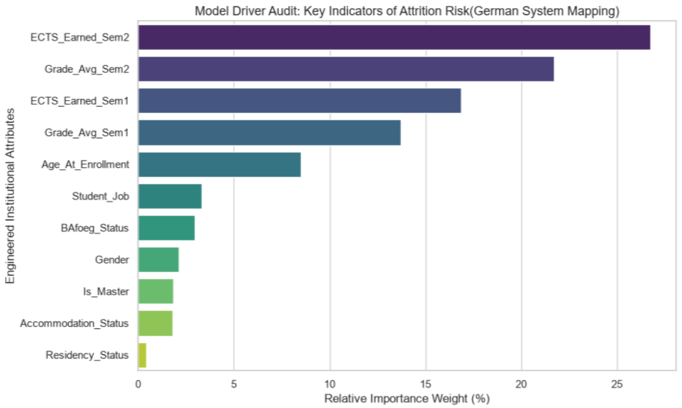
Correlation Matrix — Institutional Features



Age Density Distribution by Academic Outcome



Calibrated Confusion Matrix (Threshold-Adjusted)



Model Driver Audit — Feature Importance

Decision Boundary Re-Calibration and K-Means Clustering

Strategic Threshold Tuning

Proactive 0.35

Standard 0.50

Set a manageable margin of precision to reduce the number of missed at-risk individuals for early intervention when warning indicators appear.

Unsupervised Persona Profiling — K-Means Clustering of At-Risk Students

Acute Academic Collapse

Severe, fast-moving academic breakdown – requiring immediate peer-mentorship intervention.

Stabilized Academic Deficit

Persistent but manageable underperformance –routed to targeted academic advisory.

Latent Socioeconomic Risk

Financial stress, residential status, and accommodation stability as socio-economic impact factors –needing proactive intervention

Knowledge Base Extraction and Chunking

Raw Regulatory Text

University of Siegen examination guidelines for Bachelor’s and Master’s students (English PDF) extracted via a local text reader.

Contextual Fragmentation

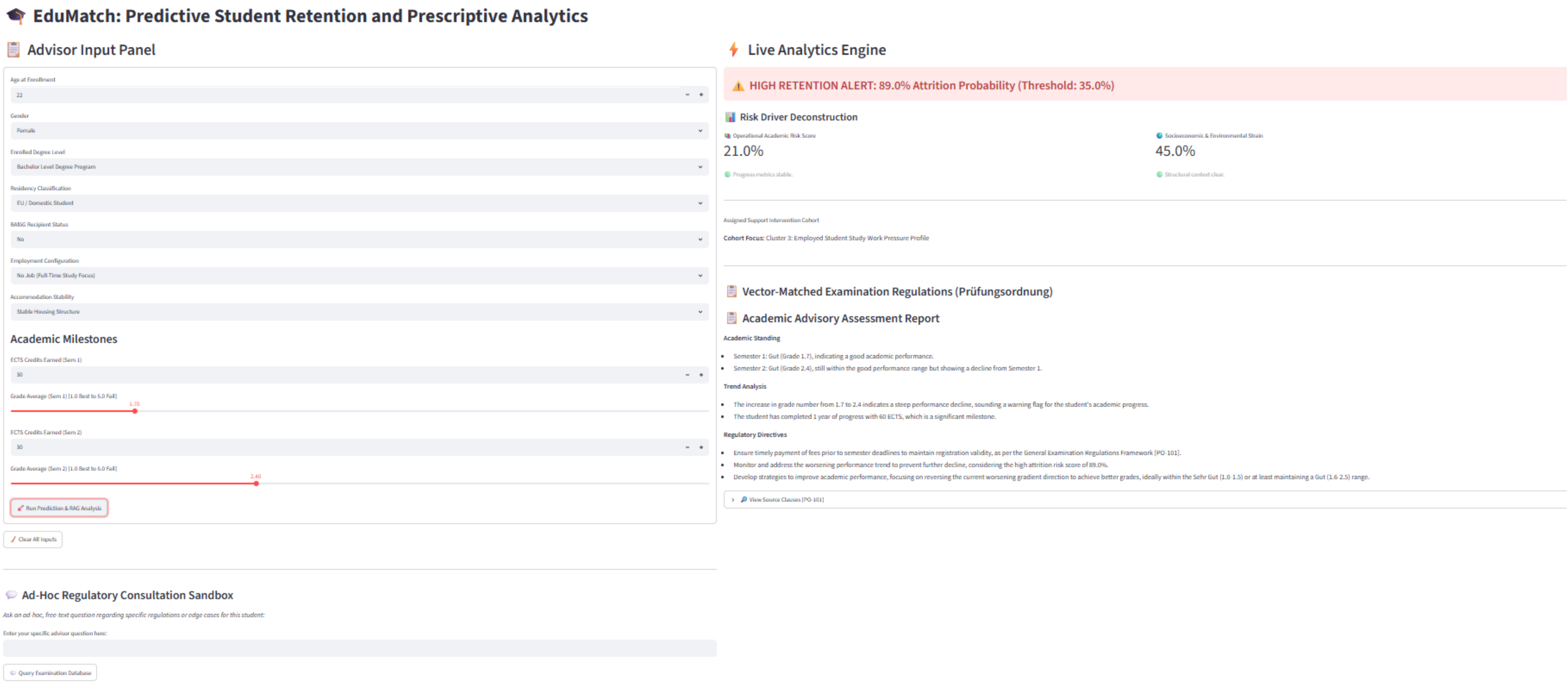
Custom parsing class cuts continuous text walls into explicit character chunks, avoiding ‘lost-in-the-middle’ bias.

Sliding Window Overlap

A character overlap window preserves continuity, so sentences and legal clauses are never split.

Local Knowledge Assembly: All outputs are consolidated into a portable local staging index (`raw_extracted_po.txt`), bypassing external cloud dependencies.

Streamlit UI Interactive Dashboard



THANK YOU FOR YOUR ATTENTION

VIELEN DANK!
ME DAASE!

I’m happy to take questions, feedback, or recommendations for my next career.